Henry Routson

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Extensive knowledge and experience in modern AI systems and Software Engineering. Currently working as an AI Engineer.

Education

The University of Melbourne
Bachelor of Science, Computing and Software Systems — *Distinction*

The University of Melbourne
Master of Computer Science, AI — *in progress, part time*

- NLP: Fine Tuning, Prompt Tuning, Transformers, Hugging Face

Ballarat Grammar

- Academic scholarship
- VCE: Software Development, Specialist & Methods Mathematics, Chemistry, Physics

Achievements

Antler VC Build Hackathon, 1st Place Jun 2026
antler.co | Event page

- Won 1st place and $7,500 prize at Antler's two-day build sprint in Sydney.
- Built auto-pipeline.dev and auto-pipeline-app.dev in 14 hours.

Experience

KPMG, AI Engineer Jul 2025 – Present
kpmg.com/au/en.html

- AI Prototype Engineer in Futures, working with LLMs and agent frameworks.
- Developed standardised codebase (34k lines) to automate prompting, parsing, validation, and deep research.
- Automated assessment of 10k pages of legislation equivalent.
- Created pipeline for SQL migration and database merging.
- Deployed Transformer models to CUDA GPUs with Hugging Face and Ollama.
- Built autonomous agent system with RLMs; multi-agent and sub-agent implementation.
- *Skills:* Azure, Ollama, FastAPI, Pydantic, DeepSeek-OCR, Hugging Face, LangChain, LangExtract, Pandas, Docker, Ray, Local model deployment, Recursive Language Models (RLM).

Antler Australia, Residency Founder (AS16) Jul 2026 – Aug 2026
antler.co/location/australia

- 80 selected from 1,600 applicants for Antler Australia's 16th inception-stage VC residency across Sydney, Melbourne, and Brisbane.
- 10-week full-time program: co-founder matching, idea validation, and pre-seed investment pathway.

Bet Right, Data Science Intern Jan 2025 – Apr 2025
linkedin/bet-right

- SQL, Snowflake, Databricks, PowerBI, Azure, DevOps.
- Improved abuse detection ML model by 15–40% using stacking.

DSCubed, DS/AI Team Vice Head Nov 2024 – Jul 2025
dscubed.org.au

- RAG, vector databases, LLM frameworks and APIs.

ARES, TypeScript Developer, University of Melbourne 2023
linkedin/aresunimelb

- 3D front-end software for rocket trajectory simulation across supersonic and subsonic speeds.

LeetCode: leetcode.com/u/HenryRoutson

Certifications

Microsoft Certified: Azure Fundamentals Apr 2025
learn.microsoft.com

Open-Source Contributions

Cached File Explorer (Rust)
github · FileExplorer/pull/47

- Ultra-fast file explorer using caching and low-level Rust.
- Fixed a poisoned thread issue in a cached file explorer implemented in Rust.

Supabase SQL Auth UI
github · flutter-auth-ui/pull/108

- Added options and documentation to the auth UI.

Projects

IT Project (JS/TS, React) dev-aia.vercel.app

- Quicksort visualisation; redesigned stack visualisation and improved code quality.

LLM Experiments github/llm_experiments

- Transformers, JAX, Keras, PyTorch; combining pre-training and supervised fine-tuning.

Auto Avail (JS/TS) github/Auto_Avail

- when2meet-style scheduler using location data to generate continuous availability periods.
- Stack: Supabase, Flutter, React Native, Postgres, pgTAP, Expo, TypeScript.

CHelp (C) github/CHelp

- Valgrind alternative: tracks dynamic allocations and catches memory issues early.

Autoheader (Rust) github/autoheader

- Automatically generates C header files.

References

Available upon request.

- **Cindy** — AI Engineer, KPMG. LinkedIn
- **Pranav** — L3 Data Scientist, NAB. LinkedIn